Q6

A) Most important phase in 2pl
    a) The most important phase in 2PL for data access during transactions is the growing phase, during which locks are acquired on the required data items before any modifications are made to them. This phase is critical to ensure that the transaction can execute without conflicting with other transactions that may also need to access the same data items.

B) When locks get released in 2pl what might cause the issue? Any fix?
    a) If locks are released prematurely in 2PL, it can cause a phenomenon known as "lost updates". This occurs when two or more transactions attempt to modify the same data item concurrently, leading to one transaction overwriting the changes made by another transaction. To fix this issue, locks should be held until the transaction completes or aborts.

C) What issue might arise if we do ™ without locks? Any fix?
    a) When performing transaction management (TM) without locks, there is a risk of conflicts between transactions that may access the same data items concurrently. This can result in inconsistencies in the data or incorrect results being returned. One way to fix this issue is to use a technique known as "optimistic concurrency control", which involves allowing transactions to proceed without locking the data, but checking for conflicts before committing the changes. If a conflict is detected, one of the transactions is rolled back and restarted.